

Calculus

HW1

65070021

นายภิธานธิราน กัศนพร มณี

Sec 1



Homework

65070021

นายภิฑิตธิ์ธนา

ทัศนพรหมลิขิต

find the limit (if it exists).

$$1. \lim_{x \rightarrow 6} (x - 2)^2$$

$$= (6 - 2)^2$$

$$= 4^2$$

$$= 16$$

~~XXXX~~

$$2. \lim_{x \rightarrow 7} (10 - x)^4$$

$$= (10 - 7)^4$$

$$= 3^4$$

$$= 81$$

~~XXXX~~

$$3. \lim_{t \rightarrow 4} \sqrt{t + 2}$$

$$= \sqrt{4 + 2}$$

$$= \sqrt{6}$$

~~XXXX~~

$$4. \lim_{y \rightarrow 4} 3|y - 1|$$

$$|y| \begin{cases} y & ; y \geq 0 \\ -y & ; y < 0 \end{cases}$$

$$|y-1| \begin{cases} y-1 & ; y-1 \geq 0 \\ -(y-1) & ; y-1 < 0 \end{cases}$$

$$= \begin{cases} y-1 & ; y \geq 1 \\ -(y-1) & ; y < 1 \end{cases}$$

$$\lim_{y \rightarrow 4} 3|y-1| = 3(4-1)$$

$$= 3(3)$$

$$= 9$$

~~XXXX~~

$$5. \lim_{t \rightarrow -2} \frac{t+2}{t^2-4}$$

$$= \lim_{t \rightarrow -2} \frac{\cancel{t+2}}{(\cancel{t+2})(t-2)}$$

$$= \lim_{t \rightarrow -2} \frac{1}{t-2}$$

$$= \frac{1}{(-2)-2} = -\frac{1}{4}$$

$$6. \lim_{t \rightarrow 3} \frac{t^2-9}{t-3}$$

$$= \lim_{t \rightarrow 3} \frac{(\cancel{t-3})(t+3)}{\cancel{t-3}}$$

$$= \lim_{t \rightarrow 3} t+3 = 3+3 = 6$$

$$7. \lim_{x \rightarrow 4} \frac{\sqrt{x-3}-1}{x-4}$$

$$= \lim_{x \rightarrow 4} \frac{\sqrt{x-3}-1}{x-4} \cdot \frac{\sqrt{x-3}+1}{\sqrt{x-3}+1}$$

$$= \lim_{x \rightarrow 4} \frac{x-3+\sqrt{x-3}-\sqrt{x-3}-1}{(x-4)(\sqrt{x-3}+1)}$$

$$= \lim_{x \rightarrow 4} \frac{\cancel{x-4}}{(\cancel{x-4})(\sqrt{x-3}+1)} = \frac{1}{\sqrt{4-3}+1} = \frac{1}{2}$$

$$8. \lim_{x \rightarrow 0} \frac{\sqrt{4+x}-2}{x}$$

$$= \lim_{x \rightarrow 0} \frac{\sqrt{4+x}-2}{x} \cdot \frac{\sqrt{4+x}+2}{\sqrt{4+x}+2}$$

$$= \lim_{x \rightarrow 0} \frac{4+x-4}{x(\sqrt{4+x}+2)} = \lim_{x \rightarrow 0} \frac{x}{x(\sqrt{4+x}+2)}$$

$$= \frac{1}{\sqrt{4+0}+2}$$

$$= \frac{1}{2+2} = \frac{1}{4}$$